Lecture 29

Existence of Analytic Solutions and Equations with Analytic Coefficients

So far, we have been considering using power series to solve differential equations, but have yet to discuss when such solutions exist and what their intervals of convergence are.

Theorem: Suppose that x_0 is an ordinary point for the equation

analytic (taylor series)

$$y'' + P(x)y' + Q(x)y = 0.$$

2nd order

Then the equation has two linearly independent analytic solutions of the form

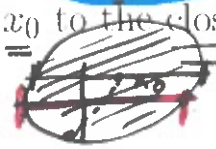
Ex: $P(x) = \frac{1}{x^2+1}$

$x^2+1 = 0 \Rightarrow x = \pm i$

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

$(x_0 - R, x_0 + R)$

for which each solutions converges at least on the interval $\boxed{|x - x_0| < R}$ where R is the distance from the center x_0 to the closest singular point of the equation in the complex plane.



Example: Consider the equation $(x^2 - 2x + 5)y'' + xy' - y = 0$. What is the minimum radius of convergence of a power series solution

1. about the ordinary point $x = 0$?

$|x - 0| < R \Rightarrow |x - 0| < 5$ x_0

where R is distance from 0 to $1 \pm 2i$

standard form:

$$y'' + \frac{x}{x^2-2x+5} y' - \frac{1}{x^2-2x+5} y = 0$$

$x^2 - 2x + 5 = 0$

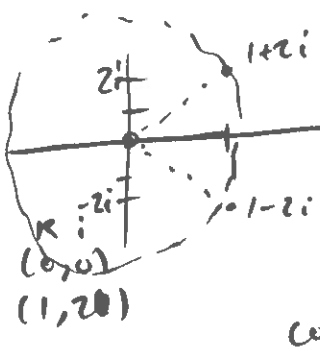
$\Rightarrow (x-1)^2 + 4 = 0$

$(x-1)^2 = -4$

$x - 1 = \pm 2i$

$\boxed{x = 1 \pm 2i}$

singular points.

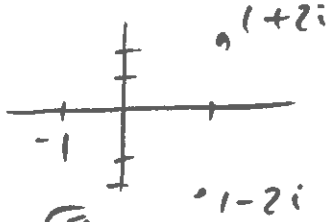


distance from real number to a imaginary conjugate pair is equal.

$R = \sqrt{(1-0)^2 + (2-0)^2} = \sqrt{1+4} = \sqrt{5}$

convergence on: $\boxed{(-5, 5)}$

2. about the ordinary point $x = -1$?



$-1 : (-1, 0)$

$1+2i : (1, 2)$

$R = \sqrt{(1-(-1))^2 + (2-0)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$

convergence on $(-1-2\sqrt{2}, -1+2\sqrt{2})$

Example: Find the first three nonzero terms of a solution centered at $\underline{\underline{x_0 = 1}}$ for the equation

$$2y'' + xy' + y = 0$$

$\Rightarrow \quad y'' + \frac{x}{2}y' + \frac{1}{2}y = 0$ standard form

no singular points.

Use $y = \sum_{n=0}^{\infty} a_n (x-1)^n$, $y' = \sum_{n=1}^{\infty} n a_n (x-1)^{n-1}$, $y'' = \sum_{n=2}^{\infty} n(n-1) a_n (x-1)^{n-2}$

xy' cannot combine x with $(x-1)^{n-1}$. we rewrite $\underset{=(x-0)}{x}$

so it is <u>centered</u> at $x=1$.

$x = (x-1) + 1 \xrightarrow{\text{method}} \begin{cases} x = a(x-1) + b & \text{solve.} \\ \Rightarrow x = ax + (b-a) \\ \Rightarrow \boxed{a = 1}, \quad b - a = 0 \\ \Rightarrow \boxed{b = 1} \end{cases}$

So $2y'' + xy' + y = 0 \Rightarrow 2y'' + [(x-1)+1]y' + y = 0$

$\Rightarrow 2\left[\sum_{n=2}^{\infty} n(n-1) a_n (x-1)^{n-2}\right] + [(x-1)+1]\left[\sum_{n=1}^{\infty} n a_n (x-1)^{n-1}\right] + \sum_{n=0}^{\infty} a_n (x-1)^n = 0$

$\Rightarrow \sum_{n=2}^{\infty} 2n(n-1) a_n (x-1)^{n-2} + \sum_{n=1}^{\infty} n a_n (x-1)^n + \sum_{n=1}^{\infty} n a_n (x-1)^{n-1} + \sum_{n=0}^{\infty} a_n (x-1)^n = 0$

down 2 up 2 down 1 up 1

$\Rightarrow \sum_{n=0}^{\infty} \overset{(1)}{2(n+2)(n+1) a_{n+2} (x-1)^n} + \sum_{n=1}^{\infty} \overset{(2)}{n a_n (x-1)^n} + \sum_{n=0}^{\infty} \overset{(3)}{(n+1) a_{n+1} (x-1)^n} + \sum_{n=0}^{\infty} \overset{(4)}{a_n (x-1)^n} = 0$

Separate $n=0$ terms:

$$\Big[\overset{(1)}{2(2)(1)a_2} + \overset{(3)}{1\cdot a_1} + \overset{(4)}{a_0}\Big] + \sum_{n=1}^{\infty}\big[2(n+2)(n+1)a_{n+2} + n a_n + (n+1)a_{n+1} + a_n\big](x-1)^n = 0$$

$$\Rightarrow \begin{cases} 4a_2 + a_1 + a_0 = 0 & (n=0) \\ 2(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} + (n+1)a_n = 0 & n \geq 1 \end{cases}$$

$$4a_2 + a_1 + a_0 = 0 \Rightarrow 4a_2 = -a_1 - a_0 \Rightarrow \boxed{a_2 = \frac{-a_1}{4} - \frac{a_0}{4}}$$

separate a_0 and a_1

$$2(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} + (n+1)a_n = 0$$

$$\Rightarrow 2(n+2)(n+1)a_{n+2} = -(n+1)(a_{n+1} + a_n)$$

$$\Rightarrow a_{n+2} = \frac{-\cancel{(n+1)}(a_{n+1}+a_n)}{2(n+2)\cancel{(n+1)}} = \frac{-(a_{n+1}+a_n)}{2(n+2)}, n \geq 1$$

$$n=1: \; a_3 = \frac{-(a_2+a_1)}{2(3)} = -\frac{a_2}{6} - \frac{a_1}{6} = -\frac{1}{6}\left[\frac{-a_1}{4} - \frac{a_0}{4}\right] - \frac{a_1}{6}$$

$$\Rightarrow \boxed{a_3 = \frac{a_0}{24} + \frac{a_1}{24} - \frac{a_1}{6} = \frac{a_0}{24} - \frac{a_1}{8}}$$

So $y = \sum_{n=0}^{\infty} a_n (x-1)^n = a_0 + a_1(x-1) + a_2(x-1)^2 + a_3(x-1)^3 + \cdots$

$$= a_0 + a_1(x-1) + \left(-\frac{a_1}{4} - \frac{a_0}{4}\right)(x-1)^2 + \left(\frac{a_0}{24} - \frac{a_1}{8}\right)(x-1)^3 + \cdots$$

$$= \boxed{\begin{aligned} & a_0\left[1 + \left(-\tfrac{1}{4}\right)(x-1)^2 + \tfrac{1}{24}(x-1)^3 + \cdots\right] \\ & + a_1\left[(x-1) + \left(-\tfrac{1}{4}\right)(x-1)^2 + \left(-\tfrac{1}{8}\right)(x-1)^3 + \cdots\right] \end{aligned}}$$

Example: Find the first four nonzero terms of a solution centered at $x_0 = -1$ for the equation

$$y'' + (3x - 1)y' - y = 0$$

use $y = \sum_{n=0}^{\infty} a_n (x+1)^n$ write $3x - 1 = 3(x+1) - 3 - 1$
$= 3(x+1) - 4$

$\Rightarrow \quad y'' + (3(x+1) - 4)y' - y = 0 \quad , x_0 = -1$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1)a_n (x+1)^{n-2} + [3(x+1) - 4] \sum_{n=1}^{\infty} n a_n (x+1)^{n-1} - \sum_{n=0}^{\infty} a_n (x+1)^n = 0$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1)a_n (x+1)^{n-2} + \sum_{n=1}^{\infty} 3n a_n (x+1)^n - \sum_{n=1}^{\infty} 4n a_n (x+1)^{n-1} - \sum_{n=0}^{\infty} a_n (x+1)^n = 0$$

up 2 down 2 up 1 down 1

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} (x+1)^n + \sum_{n=1}^{\infty} 3n a_n (x+1)^n - \sum_{n=0}^{\infty} 4(n+1)a_{n+1} (x+1)^n - \sum_{n=0}^{\infty} a_n (x+1)^n = 0$$

(1) (2) (3) (4)

Separate $n=0$ terms:

$$[2(1)a_2 - 4(1)a_1 - a_0] + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + 3n a_n - 4(n+1)a_{n+1} - a_n](x+1)^n = 0$$

(1) (2) (4)

$$\Rightarrow \begin{cases} 2a_2 - 4a_1 - a_0 = 0 & (n=0) \\ (n+2)(n+1)a_{n+2} - 4(n+1)a_{n+1} + (3n-1)a_n = 0, & n \geq 1 \end{cases}$$

$$\Rightarrow \quad 2a_2 = 4a_1 + a_0 \Rightarrow \boxed{a_2 = 2a_1 + \tfrac{1}{2}a_0} \qquad (n=0)$$

$$(n+2)(n+1)a_{n+2} - 4(n+1)a_{n+1} + (3n-1)a_n = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} = 4(n+1)a_{n+1} + (1-3n)a_n$$

$$\Rightarrow \boxed{a_{n+2} = \frac{4(n+1)a_{n+1} + (1-3n)a_n}{(n+2)(n+1)}, \quad n \geq 1}$$

recurrence relation.

$$n=1: \quad a_3 = \frac{4(2)a_2 + (1-3)a_1}{3(2)} = \frac{8a_2 - 2a_1}{6} = \frac{4}{3}a_2 - \frac{1}{3}a_1$$

$$= \frac{4}{3}\left(2a_1 + \frac{1}{2}a_0\right) - \frac{1}{3}a_1 \Rightarrow \boxed{a_3 = \frac{2}{3}a_0 + \frac{7}{3}a_1}$$

$$n=2: \quad a_4 = \frac{4(3)a_3 + (1-6)a_2}{4(3)} = a_3 - \frac{5}{12}a_2 =$$

$$= \left(\frac{2}{3}a_0 + \frac{7}{3}a_1\right) - \frac{5}{12}\left(2a_1 + \frac{1}{2}a_0\right) \Rightarrow a_4 = \frac{11}{24}a_0 + \frac{3}{2}a_1$$

$$y = \sum_{n=0}^{\infty} a_n (x+1)^n = a_0 + a_1(x+1) + a_2(x+1)^2 + a_3(x+1)^3 + a_4(x+1)^4 + \cdots$$

$$= a_0 + a_1(x+1) + \left(2a_1 + \frac{1}{2}a_0\right)(x+1)^2 + \left(\frac{2}{3}a_0 + \frac{7}{3}a_1\right)(x+1)^3 + \left(\frac{11}{24}a_0 + \frac{3}{2}a_1\right)(x+1)^4 + \cdots$$

$$= a_0\left[1 + \frac{1}{2}(x+1)^2 + \frac{2}{3}(x+1)^3 + \frac{11}{24}(x+1)^4 + \cdots\right]$$

$$+ a_1\left[(x+1) + 2(x+1)^2 + \frac{7}{3}(x+1)^3 + \frac{3}{2}(x+1)^4 + \cdots\right.$$

Example: Find the first five nonzero terms of a solution centered at $x_0 = 0$ for the IVP

not a polynomial

$$y' - e^x y = 0, \quad y(0) = 1 \quad \Rightarrow a_0 = 1$$

?

Use $y = \sum_{n=0}^{\infty} a_n x^n$ and $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$= a_0 + a_1 x + \cdots$ $\quad x=0 \Rightarrow 0$

$= 1 \quad y(0) = 1$

Substitute:

$$y' - e^x y = 0 \Rightarrow \sum_{n=1}^{\infty} n a_n x^{n-1} - \left(\sum_{n=0}^{\infty} \frac{x^n}{n!}\right)\left(\sum_{n=0}^{\infty} a_n x^n\right) = 0$$

$$\Rightarrow (a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \cdots)$$
$$-\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots\right)(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots) = 0$$

$$\Rightarrow (a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \cdots)$$
$$-(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots)$$
$$-(a_0 x + a_1 x^2 + a_2 x^3 + a_3 x^4 + \cdots)$$
$$-\left(\frac{a_0}{2} x^2 + \frac{a_1}{2} x^3 + \frac{a_2}{2} x^4 + \cdots\right)$$
$$-\left(\frac{a_0}{6} x^3 + \frac{a_1}{6} x^4 + \cdots\right)$$
$$-\left(\frac{a_0}{24} x^4 + \cdots\right) = 0$$
$$-\cdots$$

$$\Rightarrow \overset{x^0}{(a_1 - a_0)} + \overset{x^1}{(2a_2 - a_1 - a_0)}x + \left(3a_3 - a_2 - a_1 - \frac{a_0}{2}\right)x^2$$
$$+ \left(4a_4 - a_3 - a_2 - \frac{a_1}{2} - \frac{a_0}{6}\right)x^3$$
$$+ \left(5a_5 - a_4 - a_3 - \frac{a_2}{2} - \frac{a_1}{6} - \frac{a_0}{24}\right)x^4 + \cdots = 0$$

x^0: $a_1 - a_0 = 0 \Rightarrow \boxed{a_1 = a_0 (= 1)}$ $\boxed{a_0 = 1}$

condition

x^1: $2a_2 - a_1 - a_0 = 0 \Rightarrow 2a_2 = a_1 + a_0 = 2 \Rightarrow \boxed{a_2 = 1}$

x^2: $3a_3 - a_2 - a_1 - \frac{a_0}{2} = 0 \Rightarrow 3a_3 - \frac{5}{2} = 0 \Rightarrow \boxed{a_3 = \frac{5}{6}}$

x^3: $4a_4 - a_3 - a_2 - \frac{a_1}{2} - \frac{a_0}{6} = 0 \Rightarrow 4a_4 - \frac{5}{6} - 1 - \frac{1}{2} - \frac{1}{6} = 0$
$\Rightarrow 4a_4 - \frac{5}{2} = 0 \Rightarrow \boxed{a_4 = \frac{5}{8}}$

x^4: $5a_5 - a_4 - a_3 - \frac{a_2}{2} - \frac{a_1}{6} - \frac{a_0}{24} = 0 \Rightarrow 5a_5 - \frac{5}{8} - \frac{5}{6} - \frac{1}{2} - \frac{1}{6} - \frac{1}{24} = 0$
$\Rightarrow 5a_5 - \frac{3}{2} - \frac{2}{3} = 0 \Rightarrow 5a_5 - \frac{13}{6} = 0 \Rightarrow \boxed{a_5 = \frac{13}{30}}$

So $y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$
$$= \boxed{1 + x + x^2 + \frac{5}{6}x^3 + \frac{5}{8}x^4 + \frac{13}{30}x^5 + \cdots}$$

Example: We have modeled a mass-spring oscillator using a 2nd-order differential equation. In these models we assume the spring is a perfect spring and performs the same forever. However, springs in reality age and lose effectiveness over time. We can model this by adjusting the spring constant over time. A model for an aging spring can be described with the equation

$$my'' + by' + k\underbrace{e^{-\eta t}}_{\text{decay}}y = 0$$

where m is the mass, b is the damping constant, k and η are positive constants contributing to the "spring constant" $\underbrace{ke^{-\eta t}}_{\text{'new' spring constant}}$, and y is the displacement of the object from equilibrium.

— — —

Suppose that $m = 1$, $b = 2$, $k = 1$, and $\eta = 1$ with the object being displaced 1 positive unit from equilibrium and then being released. This IVP is written

$$y'' + 2y' + e^{-t}y = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

$\hookrightarrow ke^{-\eta t}, k=1, \eta=1$

Find the first four nonzero terms of a power series solution to this IVP centered at $x_0 = 0$. [How does this compare with the solution when $\eta = 0$ and we have a "perfect" spring?] [What happens in the case where there is no damping for fixed η?]

$y = \sum_{n=0}^{\infty} a_n t^n$, $\quad e^{-t} = \sum_{n=0}^{\infty} \frac{(-1)^n t^n}{n!}$, $\quad y(0)=1$, $y'(0)=0$

$\Rightarrow a_0 = 1$, $a_1 = 0$

(If $\eta=0, b=0$, $y=\cos(t)$)

shift to $n=0$

$$\Rightarrow \boxed{\sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 2\sum_{n=1}^{\infty} n a_n t^{n-1}} + \left(\sum_{n=0}^{\infty} \frac{(-1)^n t^n}{n!}\right)\left(\sum_{n=0}^{\infty} a_n t^n\right) = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \overset{①}{(n+2)(n+1)a_{n+2}} t^n + \sum_{n=0}^{\infty} \overset{②}{2(n+1)a_{n+1}} t^n + \left(\sum_{n=0}^{\infty} \overset{③}{\frac{(-1)^n t^n}{n!}}\right)\left(\sum_{n=0}^{\infty} \overset{④}{a_n t^n}\right) = 0$$

$$\overset{①}{\left[2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3 + \cdots\right]} + \overset{②}{\left[2a_1 + 4a_2 t + 6a_3 t^2 + 8a_4 t^3 + \cdots\right]}$$

$$+ \left[\underbrace{1 - t + \tfrac{1}{2}t^2 - \tfrac{1}{6}t^3 + \cdots}_{③}\right]\left[a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \cdots\right] = 0 \quad ④$$

$$\overset{①+②}{\left[(2a_2 + 2a_1) + (6a_3 + 4a_2)t + (12a_4 + 6a_3)t^2 + (20a_5 + 8a_4)t^3 + \cdots\right]}$$

$$+\left[a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \cdots\right]$$

$$-\left[a_0 t + a_1 t^2 + a_2 t^3 + \cdots\right]$$

$$+\left[\frac{a_0}{2}t^2 + \frac{a_1}{2}t^3 + \cdots\right]$$

$$-\left[\frac{a_0}{6}t^3 + \cdots\right]$$

$$+ \cdots = 0$$

$\boxed{a_0 = 1,\ a_1 = 0}$ conditions

t^0: $2a_2 + 2a_1 + a_0 = 0 \Rightarrow 2a_2 + 0 + 1 \Rightarrow \boxed{a_2 = -\frac{1}{2}}$

t^1: $6a_3 + 4a_2 + a_1 - a_0 = 0 \Rightarrow 6a_3 - 2 + 0 - 1 \Rightarrow 6a_3 = 3$ $\boxed{a_3 = \frac{1}{2}}$

t^2: $12a_4 + 6a_3 + a_2 - a_1 + \frac{a_0}{2} = 0 \Rightarrow 12a_4 + 3 - \frac{1}{2} - 0 + \frac{1}{2} \Rightarrow \boxed{a_4 = -\frac{1}{4}}$

t^3: $20a_5 + 8a_4 + a_3 - a_2 + \frac{a_1}{2} - \frac{a_0}{6} = 0 \Rightarrow 20a_5 - 2 + \frac{1}{2} + \frac{1}{2} + 0 - \frac{1}{6} = 0$

$\Rightarrow 20a_5 = \frac{7}{6} \Rightarrow \boxed{a_5 = \frac{7}{120}}$

So $y = \sum_{n=0}^{\infty} a_n t^n = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \cdots$

$$\boxed{= 1 + 0 \cdot t + \left(-\tfrac{1}{2}\right)t^2 + \tfrac{1}{2}t^3 + \left(-\tfrac{1}{4}\right)t^4 + \left(\tfrac{7}{120}\right)t^5 + \cdots}$$

$y'' + 2y' + e^{-t}y = 0$, if $\eta = 0$: $y'' + 2y' + y = 0$

$\eta = 1$

$\Rightarrow r^2 + 2r + 1 = 0$

$\Rightarrow (r+1)^2 = 0$

$\Rightarrow r = -1, -1$

So $y = c_1 e^{-t} + c_2 t e^{-t}$, $y' = -c_1 e^{-t} + c_2 e^{-t} - c_2 t e^{-t}$

$y(0) = 1 \Rightarrow c_1 + c_2(0) = 1 \rightarrow \boxed{c_1 = 1}$

$y'(0) = 0 \Rightarrow -c_1 + c_2 - c_2(0) = 0 \rightarrow \boxed{c_2 = c_1 = 1}$

$$y = e^{-t} + te^{-t} = \sum_{n=0}^{\infty} \frac{(-1)^n t^n}{n!} + t \sum_{n=0}^{\infty} \frac{(-1)^n t^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n t^n}{n!} + \sum_{n=0}^{\infty} \frac{(-1)^n t^{n+1}}{n!}$$

$$= \left(1 - \cancel{t} + \tfrac{1}{2}t^2 - \tfrac{1}{6}t^3 + \tfrac{1}{24}t^4 + \cdots\right) + \left(\cancel{t} - t^2 + \tfrac{1}{2}t^3 - \tfrac{1}{6}t^4 + \cdots\right)$$

$$= 1 - \tfrac{1}{2}t^2 + \tfrac{1}{3}t^3 - \tfrac{1}{8}t^4 + \cdots$$

The coefficients of the aging spring are larger after t^2 term. So there is more movement away from equilibrium.

If $b=0$ $\quad my'' + ke^{-\eta t}y = 0$ $\quad \sum_{n=0}^{\infty} \frac{(-\eta t)^n}{n!}$

If $m=1$, $k=1$: $\quad y'' + e^{-\eta t}y = 0$ at $x_0 = 0$.

Write out a power series solutions (4 nonzero terms).

$$\sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \left(\sum_{n=0}^{\infty} \frac{(-1)^n \eta^n t^n}{n!}\right)\left(\sum_{n=0}^{\infty} a_n t^n\right) = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} t^n + \left(\sum_{n=0}^{\infty} \frac{(-1)^n \eta^n t^n}{n!}\right)\left(\sum_{n=0}^{\infty} a_n t^n\right) = 0$$

$$(1 - \eta t + \frac{\eta^2}{2}t^2 - \frac{\eta^3}{6}t^3 \ldots)(a_0 + a_1 t + a_2 t^2 \ldots)$$

$$(2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3 + \cdots)$$

$$+ (a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \cdots)$$

$$-(\eta a_0 t + \eta a_1 t^2 + \eta a_2 t^3 + \cdots)$$

$$+(\frac{\eta^2}{2} a_0 t^2 + \frac{\eta^2}{2} a_1 t^3 + \cdots)$$

$$-(\frac{\eta^3}{6} a_0 t^3 + \cdots)$$

$$+ \cdots = 0$$

No initial conditions (separate a_0 & a_1)

$t^0: \; 2a_2 + a_0 = 0 \Rightarrow \boxed{a_2 = -\frac{1}{2}a_0}$

$t^1: \; 6a_3 + a_1 - \eta a_0 = 0 \quad \boxed{a_3 = -\frac{1}{6}a_1 + \eta\frac{1}{6}a_0}$

$t^2: \; 12a_4 + a_2 - \eta a_1 + \frac{\eta^2}{2}a_0 = 0 \Rightarrow a_4 = -\frac{1}{12}a_2 - \eta\frac{1}{12}a_1 - \eta^2\frac{1}{24}a_0$

$\Rightarrow a_4 = \frac{1}{24}a_0 - \eta^2\frac{1}{24}a_0 - \eta\frac{1}{12}a_1$

$\Rightarrow \boxed{a_4 = (1-\eta^2)\frac{1}{24}a_0 - \eta\frac{1}{12}a_1}$

$t^3: \; 20a_5 + a_3 - \eta a_2 + \frac{\eta^2}{2}a_1 - \frac{\eta^3}{6}a_0 = 0 \Rightarrow a_5 = -\frac{1}{20}a_3 + \frac{1}{20}\eta a_2$

$-\eta^2\frac{1}{40}a_1 + \eta^3\frac{1}{120}a_0$

$\Rightarrow a_5 = \frac{1}{120}a_1 - \eta\frac{1}{120}a_0 - \eta\frac{1}{40}a_0 - \eta^2\frac{1}{40}a_1 + \eta^3\frac{1}{120}a_0$

$\Rightarrow \boxed{a_5 = (\eta^3\frac{1}{120} - \eta\frac{1}{30})a_0 + (\frac{1}{120} - \eta^2\frac{1}{40})a_1}$

So $y = \sum_{n=0}^{\infty} a_n t^n = a_0 + a_1 t + (-\frac{1}{2})a_0 t^2 + a_3 t^3 + a_4 t^4 + \cdots$

$= a_0\left[1 - \frac{1}{2}t^2 + \eta\cdot\frac{1}{6}t^3 + (1-\eta^2)\frac{1}{24}t^4 + \cdots\right]$

$+ a_1\left[t + (-\frac{1}{6})a_1 t^3 + (-\frac{1}{12}\eta)t^4 + \cdots\right]$

If $\eta = 0$, $y'' + y = 0 \Rightarrow a_0\cos(t) + a_1\sin(t) = y$